FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, KanakapuraTaluk, Ramanagara District, Pin Code: 562 112

PROJECT CENTRIC LEARNING (PCL) PHASE 2

PROJECT TEAM NO.	CoEPCL-0224

PROJECT TITLE	Detection of Diabetes using Machine Learning

TEAM MEMBERS

S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.
1	23BTRCT074	YedotiPremchand	CSE*B	8919098339
2	23BTRCT011	SaiSanjay	CSE*A	8688140871
3	23BTRCT133	Tejaswanth	CSE*C	9398844671
4	23BTRCT117	SalvaAbilash	CSE*C	8951828425
5	23BTRCT112	RithikaRv	CSE*C	7358913705
6	23BTRCT111	YaswanthKumar	CSE*A	9390004799

DETAILS OF THE PROJECT GUIDE

NAME	Dr.SuriyaPrakashJ
DESIGNATION	AssociateProfessor
DEPARTMENT	Computer Science & Engineering- CSE
eMail ID	Suriyaprakash.j@jainuniversity.ac.in
Phone No.	7349456210

DOMAIN OF THE PROJECT

Machine Learning and Healthcare Analytics

Scope of the Project

1.To develop an intelligent diabetes prediction system using Machine Learning algorithms.

2.To analyze patient health data and identify diabetes at an early stage with high accuracy.

3.To compare multiple classification algorithms and identify the best-performing model.

4.To support healthcare professionals with faster and more accurate decision-making systems.

PROJECT GUIDE | **PROJECT COORDINATOR**

Existing System:-

The existing diabetes detection system mainly relies on traditional medical diagnosis methods such as blood glucose testing, laboratory investigations, physical examinations, and continuous monitoring by healthcare professionals. Patients are generally required to visit hospitals or diagnostic centers multiple times to undergo tests like fasting blood sugar tests, HbA1c tests, and oral glucose tolerance tests.

These traditional methods are often time-consuming, costly, and highly dependent on medical infrastructure and expert analysis. In many rural and underdeveloped regions, access to proper healthcare facilities and diagnostic laboratories is limited, making early diabetes detection difficult for patients.

Most existing systems focus only on manual diagnosis and do not effectively utilize intelligent data-driven technologies to analyze patient health records. As a result, hidden patterns and risk factors present in patient data may not be identified efficiently. In addition, some conventional prediction systems use only a limited number of machine learning models and may not provide high prediction accuracy.

Existing machine learning-based systems also face several limitations such as:

- Limited dataset handling capabilities
- Lower prediction accuracy
- Poor feature selection methods
- Lack of dimensionality reduction techniques
- Overfitting and underfitting problems
- Reduced performance when dealing with large healthcare datasets

Problem Identification

- Difficulty in detecting diabetes at an early stage.
- Dependence on laboratory-based testing methods.
- Limited healthcare accessibility in remote regions.
- Need for faster and more accurate prediction systems.
- High chances of delayed diagnosis and treatment.

Proposed System:-

The proposed system is an intelligent diabetes prediction framework developed using Machine Learning algorithms and Kernel Principal Component Analysis (Kernel PCA) to improve the accuracy and efficiency of early diabetes detection. The system is designed to analyze patient health records and automatically predict whether a person is diabetic or non-diabetic based on various medical and health-related parameters.

The proposed model uses a dataset containing 521 patient records with 17 important health attributes such as age, gender, BMI, glucose level, blood pressure, and other medical measurements. Before training the models, the dataset undergoes several preprocessing steps including handling missing values, normalization, feature scaling, and dimensionality reduction to improve data quality and model performance.

Kernel PCA is used as a dimensionality reduction technique to reduce the complexity of the dataset while preserving important nonlinear relationships between features. This helps improve computational efficiency, reduce noise in the dataset, and enhance the prediction capability of the machine learning models.

The system implements and compares 15 different Machine Learning classification algorithms including: AdaBoost, CatBoost, Decision Tree Classifier (DTC), Gradient Boosting Classifier (GBC), K-Nearest Neighbor (KNN), Kernel Support Vector Machine (KSVM), Linear Discriminant Analysis (LDA), Logistic Regression (LG), LightGBM (LGBM), Linear SVM (LSVM), MLP Classifier (MLPC), Naive Bayes (NB), Quadratic Discriminant Analysis (QDA), Random Forest Classifier (RFC), and XGBoost (XGB).

The dataset is divided into multiple train-test split ratios such as 80-20, 60-40, and 40-60 to evaluate the performance of each model under different testing conditions. The models are trained and tested using performance evaluation metrics such as:

- Accuracy
- Precision
- Recall
- F1-Score
- True Negative Rate

The system can also be integrated in the future with hospital management systems, electronic health records (EHR), cloud-based healthcare platforms, mobile healthcare applications, and real-time monitoring systems to support intelligent healthcare services.

Overall, the proposed system provides an efficient, reliable, and scalable solution for early diabetes prediction using advanced machine learning techniques and healthcare analytics.

No. of Modules

Sl. No.	**Name of the Module**	**Remarks**
1	Data Collection Module	Collects and organizes patient health dataset
2	Data Preprocessing Module	Handles missing values, normalization, and data cleaning
3	Feature Extraction using Kernel PCA	Reduces dimensionality and selects important features
4	Machine Learning Model Training Module	Trains 15 classification algorithms using processed data
5	Model Evaluation Module	Evaluates models using accuracy, precision, recall, and F1-score
6	Prediction & Result Analysis Module	Predicts diabetes status and compares model performance

PROJECT GUIDE **PROJECT COORDINATOR**